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8 July 1952
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72-121
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MILITARY
SPECIFICATION

SHOCK MOUNTS M-447, M-448, M-449 AND BRACKET FT-512

1. SCOPE

1.1 This specification covers three types of shock mounts, designated as Shock Mounts M-447, M-448 and M-449, and one type of bracket designated as Bracket FT-512.

1.2 Shock Mounts M-447, M-448, and M-449 are designed to support axial loads varying from 3 to 30 pound in vehicular installations. The Bracket FT-512 is designed to act as a mounting for any of the above mentioned shock mounts.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-T-111	Tape, Paper, Gummed (Sealing and Securing)

MILITARY

JAN-P-105	Packaging and Packing for Overseas Shipment-Boxes, Wood, Cleated, Plywood
JAN-P-106	Packaging and Packing for Overseas shipment-Boxes, Wood, Nailed
MIL-P-116	Preservation, Methods of
JAN-P-117	Packaging and Packing for Overseas shipment-Bags, Interior Packaging
JAN-P-120	Packaging and Packing for Overseas shipment-Cartons, Folding, Paperboard.
JAN-P-133	Packaging and Packing for Overseas shipment-Boxes, Set-up, Paperboard
MIL-L-10547	Liners, Case, Waterproof

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-129	Marking of shipments

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DRAWINGS

SIGNAL CORPS

460-1	Control Test Sampling for 5% Defective Acceptable Quality Level
460-2	Control Test Sampling Plans for 5% Defective Acceptable Quality Level
SC-D-19172	Bracket FT-512, Assembly and Details
SC-DL-27692	Shock Mount M-447
SC-DL-27699	Shock Mount M-448
SC-DL-27703	Shock Mount M-449

(Copies of specifications, standards, and drawings required by contracts in connection with specific procurement functions, may be obtained from or as directed by the Contracting Officer. Both title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following publications, of the issue in effect on date of invitation for bids, unless otherwise stated, form a part of this specification.

AMERICA SOCIETY FOR TESTING MATERIALS

D 429-39	Standard Method of Test for Adhesion of Vulcanized Rubber to Metal
D 746-44T	Tentative Method of Test for Brittle Temperature of Plastics and Elastomers.

(Applications for copies should be addressed to American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Construction.- Shock Mounts M-447, M-448, M-449 and Bracket FT-512 shall be constructed in accordance with the following drawings:

Shock Mount M-447	SC-DL-27692
Shock Mount M-448	SC-DL-27699
Shock Mount M-449	SC-DL-27703
Bracket FT-512	SC-L-19172

Capable of meeting the requirements and tests of this specification. In event of conflict between this specification and the drawings; the specification shall govern.

3.2 Quantities.- After award of the contract, the following preproduction samples shall be furnished (See 6.3.2).

24 Complete item of each type on order

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3.2.1 Compliance with specified requirements.-

3.2.1.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect.

3.2.1.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor, preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.2.1.3 When deviation from 3.2.1.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.2.2 Test data.- Preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibrations, and the like.)

3.2.3 Reference standards.- After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as referenced standards to resolve any differences of opinion regarding interpretation of specified requirements.

3.3 Service conditions.- The equipment shall be constructed to withstand long periods of storage (both before and after unpacking), and continuous operation under any combination of service conditions listed below, after which it shall be capable of meeting specified performance. The following criteria are considered the minimum requirements to insure satisfactory performance of the equipment in its intended service.

a. Extreme temperature: from -40 F to + 150 F (operating)
-60 F to + 160 F (storage)

b. Relative humidity: as high as 100 percent at specified temperature including moisture due to temperature changes.

3.4 Elastic compound.- The elastic compound used in the shock mounts shall be natural rubber or neoprene having an impact resilience of 65 to 75 percent energy recovered at 50 C (+122F). The load deflection curve of the assembled shock mounts shall be within the limits shown on the respective drawings, M-447, Fig. 1, M-448, Fig. 2, M-449, Fig. 3, when tested as required in 4.2.1.

3.5 Durability.- At the completion of one (1) hour recovery period after the test specified in 4.2.2 through 4.2.2.2, the compression side of the shock mounts shall be upright and shall conform to the applicable shock mount drawing in contour and height. There shall be no evidence of cracking or bond failure. Superficial abrasion of the rubber is permissible.

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3.6 Drift.- The estimated drift shall be not more than 0.010 inch on the basis of continued loading for one year, when tested in accordance with 4.2.3.

3.7 Low temperature.- The shock mounts when tested as specified in 4.2.4 shall stiffen not more than 300 percent of the values specified in 3.4 and shall not evidence cracking or loss of bond of the elastic compound.

3.8 Brittle temperature.- When tested in accordance with 4.2.10 through 4.2.10.4 the compound shall not become brittle at temperatures above minus 60F. (see also 6.2 through 6.2.2)

3.9 Loading.- There shall be no evidence of permanent deformation of any part, when tested in accordance with the test specified in 4.2.6.

3.10 Bonding strength.- The bonding strength shall be not less than 500 pounds per square inch, at the conclusion of the test specified in 4.2.7.

3.11 Resilience.- The resilience in percent energy recorded shall be not less than 65 percent, during or at the completion of the test specified in 4.2.8.

3.12 Corrosion.- There shall be no evidence of base metal corrosion, at the completion of the test specified in 4.2.9.

3.13 Workmanship.- All parts shall be manufactured and finished in a throughly workmanlike manner and in accordance with best commercial practice for this equipment. All dimensions, except where tolerances are given on the drawings, shall be held as close as is consistent with good practice.

4. SAMPLING INSPECTION, AND TEST PROCEDURE

4.1 Inspection, general.- Inspection shall consist of group A and B inspection as listed in Table 1 and as specified in 4.1.1 through 4.1.3.1 (See 6.3.1). Group B inspection shall be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, this order of inspection may be varied when the Government inspector considers it more practical for administrative reasons to select separate samples for group B inspection. Samples for inspection shall be selected at random by the Government inspector.

4.1.1 Group A inspection.- At the discretion of the Government inspector, Group A inspection shall be conducted by either or both of the following methods:

(a) 100-percent inspection

(b) Statistical sampling in accordance with Standard MIL-STD-105. AQL's (acceptable quality levels) shall be as specified by the contracting officer.

Order of inspection: Any order which is satisfactory to the Government inspector may be used.

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4.1.2 Group B inspection.- The AQL's for Group B shall be as shown in Table I. Sampling plans for specified AQL shall conform to drawing 400-1 and 400-2.

4.1.2.1 Rejection.- If the number of defective sample units from a lot exceeds the acceptance number specified on the drawings, the lot shall be rejected and the contractor shall take immediate action to correct the cause of failure.

4.1.2.2 Order of inspection.- Group B inspections shall be performed in any order satisfactory to the Government inspector.

TABLE I

Inspection Procedures

Inspection	Rqt. Par.	Test Par.	Preproduction	Group A % AQL	Group B % AQL
Compression	3.4	4.2.1	X		5
Durability	3.5	4.2.2	X		5
Drift	3.6	4.2.3	X		5
Low temperature	3.7	4.2.4	X		5
Brittle temperature	3.8	4.2.10	X		5
Aging of elastic compound	3.4 to 3.6	4.2.5	X		5
Loading	3.9	4.2.6	X		5
Bending strength	3.10	4.2.7	X		5
Resilience	3.11	4.2.8	X		5
Corrosion	3.12	4.2.9	X		5
Visual mechanical	3.13	4.2.11	X	X	

4.2 Tests.- The preproduction samples shall be subjected to all the tests of this specification (See Table I).

4.2.1 Compression.- The spring element assemblies shall be tested for compliance with the requirements of 3.4.

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4.2.2 Durability.- The shock mounts shall be loaded as specified in 4.2.2.1 and subjected to the test specified in 4.2.2.2. At the completion of the test the shock mounts shall meet the requirements of 3.5.

4.2.2.1 Test loading.- Four mounts installed on bracket FT-512 shall be installed on one foot centers under a dummy load. The dummy load shall weigh 25 pounds for M-447; 50 pounds for M-448; and 100 pounds for M-449. The center of gravity of the load shall be 3 inches above the top of the shock mounts.

4.2.2.2 Bounce.- With the dummy load on the shock mounts, the shock mounts shall be secured to the Vehicular Adapter Plate and placed on the table of the Package Tester, as manufactured by the L.A.B. Corporation, Summit, New Jersey, or equal. The plate shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at speeds of 285 r.p.m. ± 1 percent, for a total of 3 hours. The adapter plate shall be rotated through 90 degrees, in the same direction, at the end of each 3/4-hour period.

4.2.3 Drift.- The shock mounts shall be subjected to minimum rated load for one (1) hour and the rate of change in deflection shall be measured under continued loading. Readings shall then be taken every 24 hours for 20 days. A curve shall be plotted on longarithmic (log-log) paper and the estimated yearly drift rate determined. The temperature equilibrium shall be assured by taking all subsequent readings at the temperature of the first reading ± 1 F. At the completion of the test the shock mounts shall meet the requirements of 3.6.

4.2.4 Low temperature.- The shock mounts shall be maintained at minus (-) 40F for one hundred (100) hours. At the end of the one hundred (100) hour period and at minus (-) 40F an axial load deflection curve to 600 pounds, in accordance with Figure 1 for M-447, Figure 2 for M-448, and Figure 3 for M-449 shall apply. The shock mounts shall meet the requirements of 3.7.

4.2.5 Aging of elastic compound.- Two assembled shock mounts shall be subjected to any one of the tests of 4.2.5.1 through 4.2.5.4. At the completion of the tests the shock mounts shall be capable of meeting the requirements of 3.4, 3.5 and 3.6.

4.2.5.1 For 96 hours in air at minus (-) 60F.

4.2.5.2 For 96 hours in air at plus (+) 150 F.

4.2.5.3 For 96 hours at plus (+) 150 F and 95 percent relative humidity.

4.2.5.4 For 96 hours in an oxygen bomb at 150 F and at a pressure of 300 pounds per square inch.

4.2.6 Loading.- Shock mounts installed in Bracket FT-512 shall be adjusted to an axial and radial load of 1500 pounds applied for one minute. Upon removal of the load the shock mounts shall meet the requirements of 3.9. The test shall be repeated with the axial load applied in the opposite direction. At the completion of the test the shock mounts shall conform to 3.9.

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VARIABLE TRANSFORMER (SEE TABLE III)

4.2.7 Bonding strength.- One complete shock mount shall be prepared using the same steel and techniques used in manufacture of the mounts. Strength of bond of the elastic compound to metal shall be tested as prescribed in American Society of Testing Materials, Standard D429. At the completion of the test the mounts shall meet the requirements of 3.10.

4.2.8 Resilience.- Appropriate test samples of the elastic compound shall be prepared and tested at 50C using the Lupke or Dunlop pendulum tester. The resilience shall meet the requirements of 3.11.

4.2.9 Corrosion.- Shock Mounts M-447, M-448, M-449 and Base FT-512 shall be given a 100 hour of exposure to the salt spray test as prescribed in Specification QQ-M-151. At the conclusion of the test the components shall meet the requirements of 3.12.

4.2.10 Brittle temperature of elastic compound.- Appropriate test specimens of the elastic compound shall be tested in accordance with 4.2.10.1 through 4.2.10.4 (see also method for brittle temperature of plastics and elastomers, American Society for Testing Materials, ASTM-D-746-44T.) At the completion of the test of 4.2.10.1 through 4.2.10.4, the compound shall meet the requirements of 3.8.

4.2.10.1 Test specimens.- The test specimens shall be 1.50 ± .05 inches in length, 0.250 ± .015 inches in width, and .075 ± .010 inches in thickness. Other values of specimen thickness may be used provided it can be shown that they give equivalent results for the material being tested.

4.2.10.2 Heat transfer medium.-Any heat transfer medium which remains fluid at the test temperature and which will not appreciably affect the materials being tested may be used.

4.2.10.3 Specimen clamp and striking member.- The specimen clamp, designed to hold the specimen as a cantilever beam, and the striking edge shall be capable of motion relative to each other at a constant linear speed of 6.5±.5 feet per second at and during impact. The distance between the center line of the striking edge and the clamp shall be .310 ± .01 inches at impact. The striking edge shall have a radius of .062 ± .005 inches.

4.2.10.4 Liquid heat transfer.- The test specimen shall be mounted in the apparatus and submerged for 2.5 ± .5 min at the test temperature. The temperature shall be recorded and the impact blow delivered.

4.2.10.5 Air heat transfer.- The test specimens shall be inserted in the clamp using forceps to avoid warming the specimen with the hands, and after closing the chamber, the impact blow shall not be initiated until 5 minutes after the air in the test chamber has returned to its equilibrium value.

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4.2.11 Visual and mechanical inspection.- Parts and equipment shall be inspected for workmanship, mechanical fit, and miscellaneous defects. Clearances dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required. Examination also may be made for other visual or mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government inspector as major or minor in accordance with the definitions of Standard MIL-STD-105.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 For domestic shipment.- Shock Mounts M-447, M-448, M-449 and Bracket FT-512 shall be preserved and packaged in accordance with best commercial practice. The equipment shall be adequately cushioned, blocked, and braced, using suitable materials and containers as required, in such a manner as to afford maximum protection from hazards normally encountered in storage and transit.

5.1.2 For overseas shipment.- Preserve and package Shock Mounts M-447, M-448, M-449 and Bracket FT-512 in accordance with procedures specified for the designated methods as prescribed in Specification MIL-P-116 and further described in the following paragraphs.

5.1.2.1 Package each item Method 10-3, as follows: Cushion each item by wrapping in flexible corrugated paper. Secure cushioning with Kraft gummed tape conforming to Specification UU-T-111. Inclose each cushioned item within a close fitting type II bag, conforming to the requirements of Specification JAN-P-117, for Class B. Properly seal closure.

5.1.2.2 Intermediate package.- Packaged items bearing the same stock number, shall be placed together within a close-fitting, folding carton or set up box, complying with Specification JAN-P-120 or JAN-P-133 respectively. Seal entire closure with gummed Kraft tape, conforming to the requirements of Specification UU-T-111, or other suitable seal. The net weight of contents shall not exceed 5 pounds.

5.2 Packing.-

5.2.1 For domestic shipment.- Shock Mounts and Brackets, packaged as specified in 5.1.1, shall be packed in substantial containers of the type, size, and kind commonly used for the purpose, in such a manner as to insure acceptance at and safe delivery to the designated point by common or other carrier at the lowest rate.

5.2.2 For export shipment.- The Shock Mounts and Brackets, packaged as specified in 5.1.2.2 shall be packed in a cleated plywood or nailed wood box, conforming to the requirements of JAN-P-105 and JAN-P-106 respectively. Fabricate the boxes to fit contents snugly. Unless otherwise specified, shipping containers shall be lined with a sealed waterproof case liner conforming to Specification MIL-L-10547. The approximate gross weight of any exterior shipping container shall not exceed 150 pounds. Close and strap the shipping container as specified in the appendix of the applicable box applicable box specification.

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5.3 Marking for shipment.- Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of MIL-STD-129.

On overseas shipping containers, the Signal Corps Insignia shall be placed on the lower left hand corner of the side containing contract data.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type required.

(c) Whether domestic or export shipment is required.

(d) Samples required. (See 3.2.)

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Acceptable quality level (See 4.1.1)

6.2 Definitions.-

6.2.1 Failure.- Failure is defined as the division of the test specimen into two or more completely separated pieces. The occurrence of fissures or cracks does not constitute brittle type failure.

6.2.2 Brittle temperature.- Brittle temperature is defined as the lowest temperature of non-failure of five consecutive specimens under the specified test conditions.

6.3 Contractual requirements.- It is recommended that the following be included in the contract:

6.3.1 Additional testing.- It should be understood that additional inspection and non-destructive testing of equipment may be performed by the Government inspector when considered necessary to determine compliance with this specification and other applicable documents. The Government inspector, therefore, may withdraw materials, parts, subassemblies, or components temporarily from production for such inspection and testing, performed either at a Government laboratory or the contractor's plant.

6.3.2 Waiver of preproduction samples.- When items for which preproduction samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should include a statement indicating that the specified samples have been submitted and approved on the current contract; production of the item under the new order will be concurrent with, or follow without interruption, production on the current order; and current production is in strict compliance with specified requirements for the new equipment, or describing any deviations authorized on the current order. Waiver of preproduction may be contingent on renegotiation of the contract to reflect the saving in cost to the contractor.

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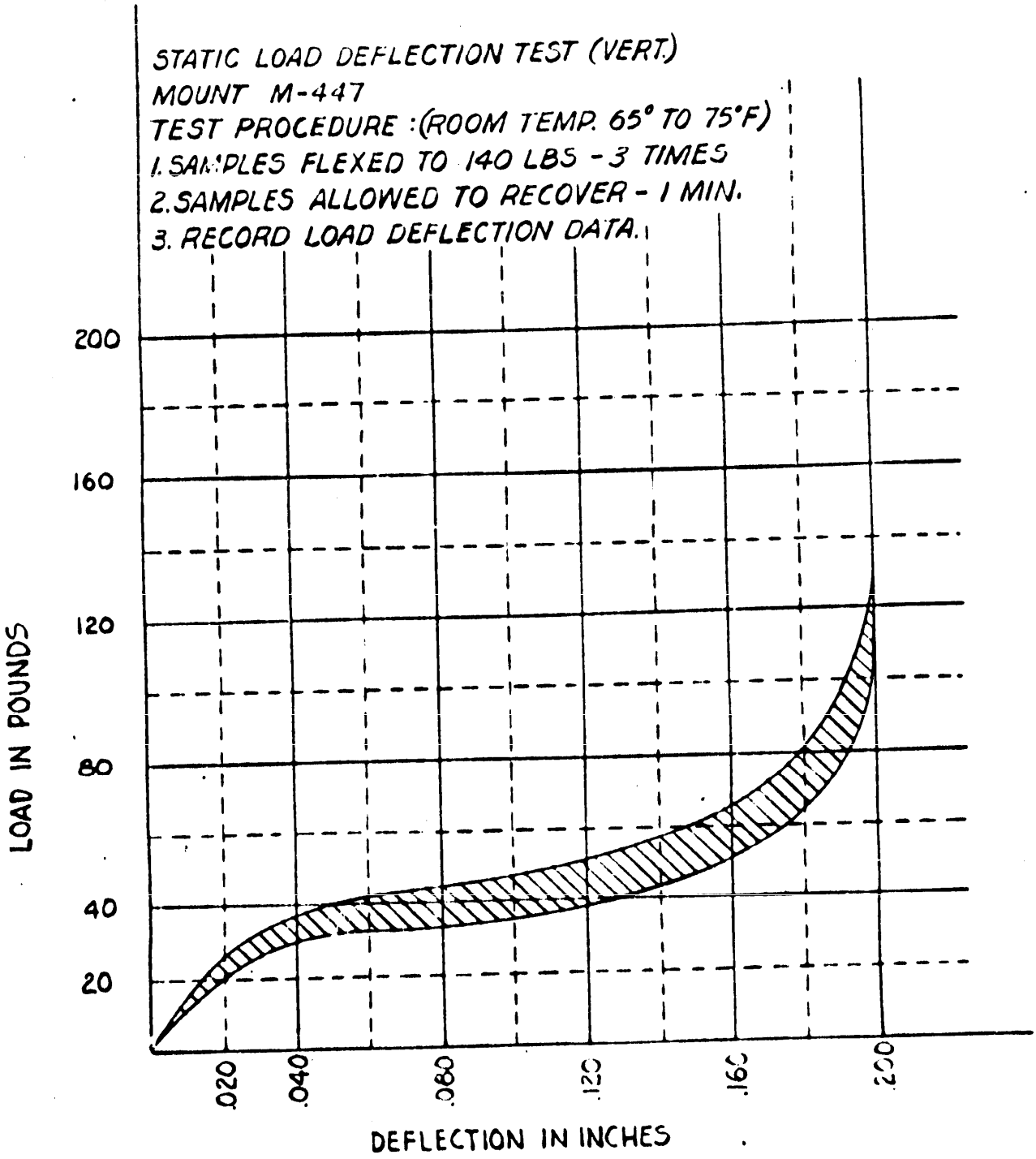


FIG. 1

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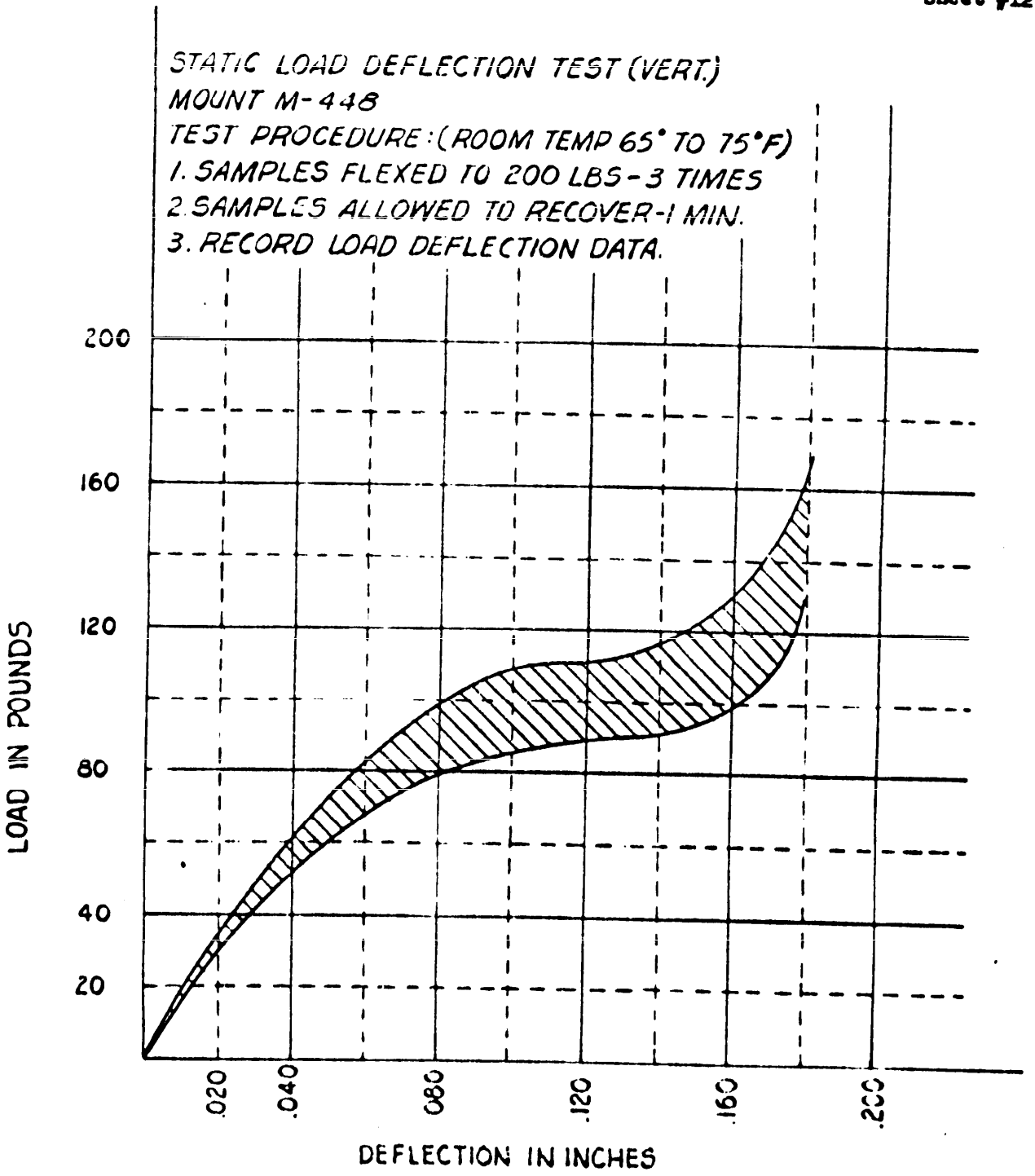


FIG. 2

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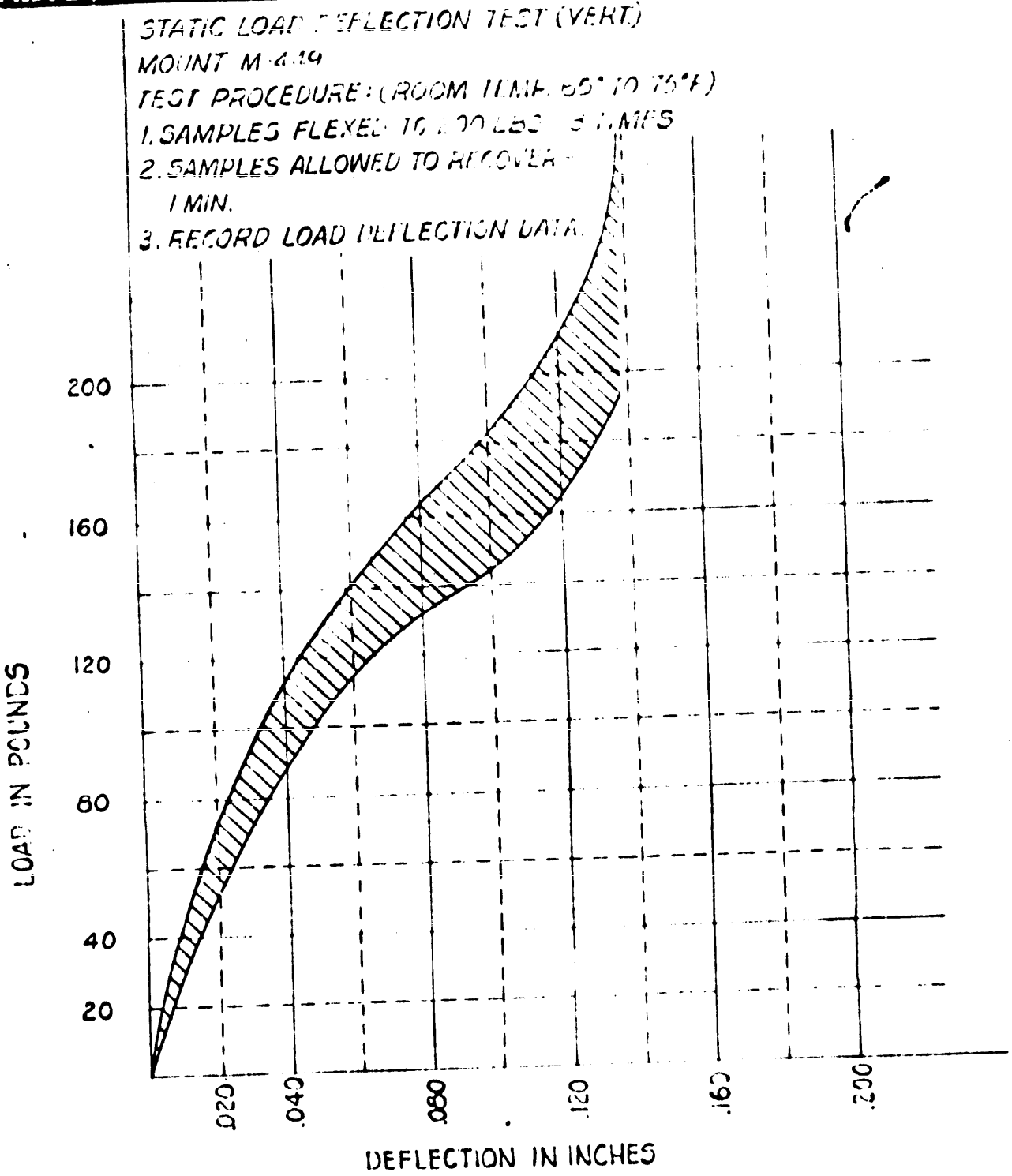


FIG 3

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